

Sol Quick Reference

Quick reference for ASU's Sol supercomputer - the solx CLI, Slurm job routing, safe automation, storage, and compute-node access.

v1.0.2



ACCESS / KNOW YOUR ACCESS FIRST

Your account decides which partitions and QOS you may use. Check it before choosing a route:

```
sacctmgr -n show assoc user=$USER format=Account,Partition,QOS
myfairshare # read RealFairShare; a very low score means longer queue waits
```

ROUTING / PARTITIONS

GPUs live in **htc**, **public**, **general**, **lightwork**, and **arm**. The deciding questions are how long the job runs, whether it may be preempted, and which accelerator it needs.

Partition	Wall limit	Hardware / best use
htc	4 h	Default for short CPU and GPU work; large accelerator pool.
public	7 days	Non-preemptable CPU/GPU jobs that need more than 4 h.
general	14 days	Privately-owned CPU/GPU nodes via private or grp_* QOS.
lightwork	1 day	Light development, compilation, bulk I/O, and vscode ; max 8 cores.
highmem	7 days	Memory-heavy CPU work, up to 2 TB; normal public use may have a lower cap.
arm	7 days	ARM/aarch64 nodes with Grace Hopper (gh200) GPUs.
fpga	7 days	FPGA, Vector Engine, and special accelerator workloads.

ACCESS / QOS OVERLAYS

QOS	Wall cap	Use it for
public	Partition limit	Default, non-preemptable access to public resources.
debug	15 min	High-priority smoke tests; one running job and two submitted jobs per user.
private	Partition limit	Preemptible access to idle buy-in nodes; owners may cancel the job.
grp_*	Site/group limit	Your group's owned nodes, when your account provides the QOS.
long	14 days	Approved long batch jobs on public or highmem ; not for interactive jobs.
class	1 day	Course accounts; per-user CPU, memory, GPU, and job-count caps.

DECISION / ROUTING IN ONE GLANCE

Routing: up to 4 h -> **htc**; up to 15 min and urgent -> **-p htc -q debug**; more than 4 h -> **public**; more than 4 h and preemption is acceptable -> **-p general -q private**. Validate unusual combinations with **sbatch --test-only**.

WORKFLOW / EVERYDAY SOLX WORKFLOW

```
solx init # create ~/.config/solx/config.toml
solx config edit # define job templates and [keep] paths
solx job start debug -n # preview the salloc command
solx job start debug # allocate; waits for the grant and prints the job ID
solx job jump # open a shell on the compute node
solx job time # show remaining wall-time
exit # leave the shell; the allocation keeps running
solx job stop # cancel when done; prompts before acting
```

job also accepts **jobs**; **job list** accepts **ls**; and **solx jump** is an alias for **solx job jump**.

COMMANDS / COMPLETE SOLX COMMAND MAP

Command	Purpose / important options
solx init [-f]	Write starter config; -f / -y overwrites.
solx job list	List your jobs (squeue --me).
solx job start [TEMPLATE]	Start an interactive allocation; -n , --timeout , and --passthrough .
solx job jump [JOBID]	Attach with srun --pty ; -q hides nesting/selection notes.
solx job time [JOBID]	Print remaining time in D-HH:MM:SS.
solx job stop [JOBID]	Cancel a job; -n previews and -y skips confirmation.
solx keep	Renew flagged scratch files selected by [keep] ; see below.
solx config show / edit	Inspect resolved config or open it in \$EDITOR .
solx completions bash zsh fish	Emit a static shell-completion script.
solx cheatsheet	Print this quick reference as text.
solx version / help	Aliases of --version / --help .

SAFETY / OUTPUT, TARGETING, AND SAFETY RULES

Situation	Rule
Human terminal	Data commands print aligned text.
Pipe or agent	Output auto-switches to JSON; --json forces it. Put --json before job start .
Destructive command	job stop and keep show a plan and prompt; -n previews, -y confirms.
Non-interactive session	A command that needs confirmation refuses instead of hanging.
Missing job ID	Inside an allocation, use \$SLURM_JOB_ID ; on login, time / jump pick the most recent job.
Multiple jobs on login	stop refuses to guess; pass the job ID.
Jump from another job	solx job jump <JOBID> safely targets that allocation and warns about nesting; use -q to silence the note.

```
solx --json job list | jq '.[].job_id'
solx job start gpu --timeout 20m -- --mem=128G # last salloc flag wins
solx job stop 12345 -n # preview, never cancel
```

SCRATCH / SOLX KEEP - BOUNDED SCRATCH RENEWAL

`solx keep` touches only paths that are both in Sol's warning CSVs and matched by the config's `[keep]` include/exclude rules. It never walks all of `/scratch` blindly.

```
solx keep --dry-run -v      # inspect the full plan first
solx keep                  # execute with a confirmation prompt
solx keep --stage pending  # only the most urgent warning CSV
solx --json keep -n        # machine-readable counts and a capped sample
```

Other controls: `--stage all|inactive|over90|pending`, `--csv-dir DIR`, `-j N` workers, `-v`, `-n`, and `-y`. Run large renewals on the DTN (`ssh soldtn`), a compute node, or a short batch job - not on a throttled login node.

BATCH / SLURM BASICS

```
sbatch job.sh              # submit a batch script
squeue --me                # your jobs (human alias: myjobs)
scancel <jobid>            # cancel
scontrol show job <jobid>  # full detail
sbatch --test-only job.sh  # validate without submitting
interactive                # quick shell; defaults to htc/public, 1 core, 4 h
```

Minimal `#SBATCH` header (time format is `D-HH:MM:SS`):

```
#!/bin/bash
#SBATCH -p htc
#SBATCH -q public
#SBATCH -t 0-04:00:00
#SBATCH -c 8
#SBATCH --gres=gpu:a100:1
#SBATCH --mem=64G
#SBATCH -o slurm.%j.out
```

Start from `/packages/public/sol-sbatch-templates/templates/` when a supplied template fits. Use `solx job start` for interactive work and `sbatch` for unattended batch work.

TRANSLATE / SOLX AND RAW SLURM EQUIVALENTS

	Raw Slurm
<code>solx job start [TEMPLATE]</code>	<code>salloc / interactive</code> with template flags.
<code>solx job jump [JOBID]</code>	<code>srun --jobid=ID --overlap --pty \$SHELL</code> .
<code>solx job list</code>	<code>squeue --me</code> .
<code>solx job time [JOBID]</code>	<code>squeue -h -j ID -o %L</code> .
<code>solx job stop -y ID</code>	<code>scancel ID</code> .

DIAGNOSE / JOB STUCK PENDING? DIAGNOSE BEFORE REROUTING

```
squeue --me -t PD -O "JobID,Reason:50,StartTime"  # reason and estimated start
scontrol show job <id>                          # all fields for one job
```

Reason	Response
Priority with low fairshare	Report the ETA and wait; resubmitting does not improve priority.
ReqNodeNotAvail	Check whether the requested node is reserved, drained, or down.
Resources	Right-size or reroute if another eligible partition starts sooner.

Only capacity-bound jobs benefit from rerouting. If a reroute helps, preserve the accrued queue priority with `scontrol update job <id> Partition=... QOS=...` instead of canceling and resubmitting.

AUTOMATION / HUMAN WRAPPERS AND AGENT-PARSEABLE COMMANDS

You want	Human command	Parse this
Fairshare / priority	<code>myfairshare</code>	<code>myfairshare -> RealFairShare</code> .
Scratch quota	<code>-</code>	<code>beegfs-ctl --getquota --uid \$USER</code> .
Current jobs	<code>myjobs</code>	<code>squeue --me -O JobID,State,Reason</code> .
Pending start estimate	<code>thisjob ID</code>	<code>scontrol show job ID -> StartTime</code> .
Finished-job efficiency	<code>seff ID</code>	<code>seff ID</code> .
Partition capacity	<code>showparts</code>	<code>sinfo -h -o "%P %a %l %D %t"</code> .
Free GPUs	<code>showgpus</code>	<code>sinfo -h -O "Partition,StateLong,Gres,GresUsed"</code> .

`my*` / `show*` tools are colorized for people. Agents should prefer native Slurm fields or `solx --json`; free GPUs equal `Gres` minus `GresUsed`.

CONNECT / REACH A COMPUTE-NODE SERVICE FROM YOUR LAPTOP

```
# On a Sol login node: register a VS Code tunnel on lightwork.
vscode

# On your laptop: forward Jupyter on compute node $NODE, port 8888.
ssh -N -L 8888:localhost:8888 -J $USER@login.sol.rc.asu.edu $USER@$NODE
```

Get `$NODE` from the `NODELIST` column in `squeue --me`. Bind services to `localhost`, never `0.0.0.0`, on shared nodes.

STORAGE / STORAGE AND HEAVY I/O

Path	Use	Policy
<code>/scratch/\$USER</code>	Datasets, caches, checkpoints, outputs.	Temporary, not backed up; inactive files are purged.
<code>/home/\$USER</code>	Code, config, and small user installs.	Persistent, backed up, small quota.

```
export HF_HOME=/scratch/$USER/.cache/huggingface
export UV_CACHE_DIR=/scratch/$USER/.cache/uv
```

Use the DTN (`ssh soldtn`) for bulk transfer and metadata-heavy I/O. Use a compute node or batch job for compute. Keep heavy work off login nodes.

REMEMBER / FIVE RULES TO REMEMBER

Rule	Default
Preview changes	Use <code>-n / --dry-run</code> before <code>job stop</code> , <code>keep</code> , or allocation changes.
Parse output	Use <code>solx --json ...</code> or native Slurm fields; do not scrape colorized wrappers.
Route short jobs	Use <code>htc</code> for work up to 4 h, including GPU jobs.
Protect the login node	Move compute and metadata-heavy I/O to a compute node, batch job, or DTN.
Preserve queue priority	Diagnose <code>PENDING</code> ; update a viable route in place instead of resubmitting.